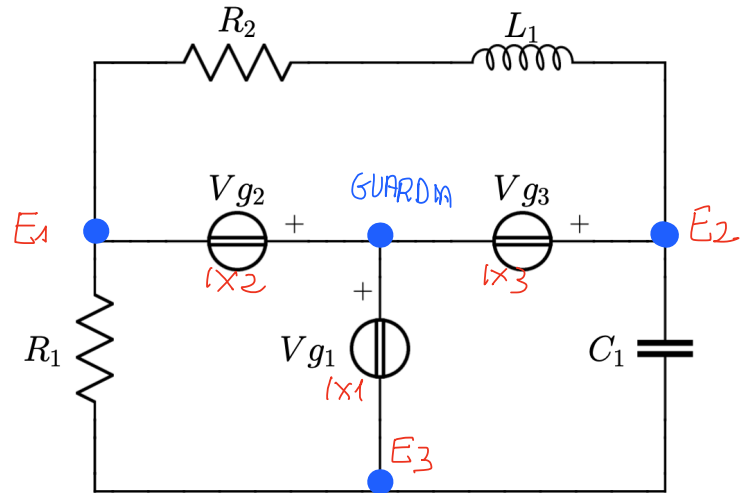


Serie NODI

domenica 3 agosto 2025 13:55

Risolvere il circuito in figura



$$\begin{aligned} R_1 &= 1 \\ v_{g1}(t) &= \cos(t + \pi) \\ C_1 &= 1 \\ v_{g2}(t) &= \cos(t + \frac{\pi}{2}) \\ v_{g3}(t) &= 2 \cos(t) \\ R_2 &= 2 \\ L_1 &= 1 \end{aligned}$$

α (radian ti)	α (gradi)	$\text{sen}\alpha$	$\text{cos}\alpha$	$\text{tang}\alpha$	$\text{cosec}\alpha$	$\text{sec}\alpha$	$\text{cotan}\alpha$
0	0°	0	1	0	N.E.	1	N.E.
$\pi/6$	30°	1/2	$\sqrt{3}/2$	$\sqrt{3}/3$	2	$2/\sqrt{3}$	$3/\sqrt{3}$
$\pi/4$	45°	$\sqrt{2}/2$	$\sqrt{2}/2$	1	$2/\sqrt{2}$	$2/\sqrt{2}$	1
$\pi/3$	60°	$\sqrt{3}/2$	1/2	$\sqrt{3}$	$2/\sqrt{3}$	2	$1/\sqrt{3}$
$\pi/2$	90°	1	0	N.E.	1	N.E.	0
$2/3\pi$	120°	$\sqrt{3}/2$	-1/2	$-\sqrt{3}$	$2/\sqrt{3}$	-2	$-1/\sqrt{3}$
$3/4\pi$	135°	$\sqrt{2}/2$	$-\sqrt{2}/2$	-1	$2/\sqrt{2}$	$-2/\sqrt{2}$	-1
$5/6\pi$	150°	1/2	$-\sqrt{3}/2$	$-\sqrt{3}/3$	2	$-2/\sqrt{3}$	$-3/\sqrt{3}$
π	180°	0	-1	0	N.E.	-1	N.E.
$7/6\pi$	210°	-1/2	$-\sqrt{3}/2$	$\sqrt{3}/3$	-2	$-2/\sqrt{3}$	$3/\sqrt{3}$
$5/4\pi$	225°	$-\sqrt{2}/2$	$-\sqrt{2}/2$	1	$-2/\sqrt{2}$	$-2/\sqrt{2}$	1
$4/3\pi$	240°	$-\sqrt{3}/2$	-1/2	$\sqrt{3}$	$-2/\sqrt{3}$	-2	$1/\sqrt{3}$
$3/2\pi$	270°	-1	0	N.E.	-1	N.E.	0
$5/3\pi$	300°	$-\sqrt{3}/2$	1/2	$-\sqrt{3}$	$-2/\sqrt{3}$	2	$-1/\sqrt{3}$
$7/4\pi$	315°	$-\sqrt{2}/2$	$\sqrt{2}/2$	-1	$-2/\sqrt{2}$	$2/\sqrt{2}$	-1
$11/6\pi$	330°	-1/2	$\sqrt{3}/2$	$-\sqrt{3}/3$	-2	$2/\sqrt{3}$	$-3/\sqrt{3}$
2π	360°	0	1	0	N.E.	1	N.E.

CALCOLIAMO I FASORI

$$V_{g1}(t) = \cos(t + \pi)$$

$$V_{g1} = 1 \quad \omega = 1 \quad \varphi = \pi \rightarrow V_{g1} = \cos \tilde{t} + j \sin \tilde{t} = -1 \quad \checkmark$$

$$V_{g2}(t) = \cos(t + \frac{\pi}{2}) \rightarrow V_{g2} = \cos \frac{\tilde{t}}{2} + j \sin \frac{\tilde{t}}{2} = j \quad \checkmark$$

$$V_{g3}(t) = 2 \cos(t) \rightarrow V_{g3} = 2 \cdot \cos \tilde{t} + j \sin \tilde{t} = 2 \quad \checkmark$$

RISOLVIAMO LA SERIE:

$$Z_a = R_2 + j\omega L_1 = 2 + j$$

$$Y_a = \frac{1}{Z_a} = \frac{1}{2+j} = \frac{2-j}{5} = \frac{2}{5} - \frac{j}{5}$$

$$\left[\frac{1}{\omega} + Y_a \quad -Y_a \quad -1 \right] / (E_1) \quad / (x_2)$$

$$\begin{bmatrix} \frac{1}{R_1} & -Y_a & -\frac{1}{R_1} \\ -Y_a & j\omega C_1 + Y_a & -j\omega C_1 \\ -\frac{1}{R_1} & -j\omega C_1 & j\omega C_1 + \frac{1}{R_1} \end{bmatrix} \begin{pmatrix} E_1 \\ E_2 \\ E_3 \end{pmatrix} = \begin{pmatrix} -I_{X2} \\ I_{X3} \\ -I_{X1} \end{pmatrix}$$

$$\begin{cases} \left(\frac{1}{R_1} + Y_a\right)E_1 - Y_a E_2 + \left(-\frac{1}{R_1}\right)E_3 = -I_{X2} \\ -Y_a E_1 + (j\omega C_1 + Y_a)E_2 + (-j\omega C_1)E_3 = I_{X3} \\ -\frac{1}{R_1}E_1 + (-j\omega C_1)E_2 + \left(j\omega C_1 + \frac{1}{R_1}\right)E_3 = -I_{X1} \\ E_3 = -V_{g1} \\ E_2 = V_{g3} \\ E_1 = -V_{g2} \end{cases}$$

Risolviamo questo sistema

$$\begin{cases} \left(1 + \frac{2}{5} - \frac{1}{5}j\right)E_1 + \left(-\frac{2}{5} + \frac{1}{5}j\right)E_2 - E_3 = -I_{X2} \\ \left(-\frac{2}{5} - \frac{1}{5}j\right)E_1 + \left(j + \frac{2}{5} - \frac{1}{5}j\right)E_2 + (-j)E_3 = I_{X3} \\ -E_1 - jE_2 + (j+1)E_3 = -I_{X1} \\ E_3 = 1 \\ E_2 = 2 \\ E_1 = -5 \end{cases}$$

Sostituisco

$$\left(-5 - \frac{2}{5}j - \frac{1}{5} - \frac{4}{5} + \frac{2}{5}j - 1\right) = -I_{X2}$$

$$\rightarrow \begin{cases} I_{X2} = 5 + 2 \\ E_3 = 1 \end{cases}$$

$$\begin{cases} \frac{2}{5}j + \frac{1}{5} + 2j + \frac{4}{5} - \frac{2}{5}j - j = 1 \times 3 \\ j - 2j + j + 1 = -1 \times 1 \end{cases}$$

$$\begin{cases} 1 \times 3 = 3 + 1 & E2 = 2 \\ 1 \times 1 = -1 & E1 = -5 \end{cases}$$

BILANCIO DI POTENZA

$$VR1 = E1 - E3 = -5 - 1$$

$$VC1 = E2 - E3 = 2 - 1 = 1$$

$$VR2 + VL1 = E2 - E1 = 2 + 5$$

$$VL1 = j\omega L1 \cdot IL1 = j$$

$$VR2 = R2 \cdot IR2 = 2$$

$$Vg1 = -1 \text{ (NOTO)}$$

$$Vg2 = j \text{ (NOTO)}$$

$$Vg3 = 2 \text{ (NOTO)}$$

$$IR1 = \frac{VR1}{R1} = \frac{-5-1}{1} = -5-1$$

$$IC1 = \frac{VC1}{\frac{1}{j\omega C1}} = \frac{1}{\frac{1}{5}} = j$$

$$IR2 = IL1 = \frac{VR2 + VL1}{Za} = \frac{2+j}{2+j} = 1$$

$$Ig1 = Ix1 = -1$$

$$Ig2 = Ix2 = j+2$$

$$Ig3 = Ix3 = j+1$$

$$PcVg1 = \frac{1}{2} Vg1 Ig1^* = \frac{1}{2} (-1) (-1) = \frac{1}{2}$$

$$PcVg2 = \frac{1}{2} Vg2 Ig2^* = \frac{1}{2} j (2-j) = \frac{2j+1}{2} = j + \frac{1}{2}$$

$$PcVg3 = \frac{1}{2} Vg3 Ig3^* = \frac{1}{2} 2 (1-j) = 1-j$$

$$\rightarrow \frac{1}{2} + \frac{1}{2} + 1 = 1 + 1$$

$$PAR1 = \frac{1}{2} R1 |IR1|^2 = \frac{1}{2} 2 = 1$$

$$PAR2 = \frac{1}{2} R2 |IR2|^2 = \frac{1}{2} 2 (1) = 1$$

$$2 = 2$$



$$QL1 = \frac{1}{2} \omega L1 |IL1|^2 = \frac{1}{2} 1 1 = \frac{1}{2}$$

$$a_{c1} = -\frac{1}{2} w_{c1} |v_{c1}|^2 = -\frac{1}{2} \cdot 1 \cdot 1 = -\frac{1}{2}$$

$$\rightarrow \frac{1}{2} - \frac{1}{2} = 1 - 1$$

$$PR_{g1} = IM(PCV_{g1}) = 0$$

$$PR_{g2} = IM(PCV_{g2}) = 1$$

$$PR_{g2} = IM(PCV_{g3}) = -1$$

$$0 = 0$$

